Response to Editorial and Reviewer Comments

Information Circulation and External-Flow Suppression under Finite Capacity

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Dear Editor,

Please find below our point-by-point responses. The manuscript has been revised to state the stochastic setting explicitly, explain the origin of the time dependence, derive the information balances needed for the analysis, and define the graph and dependency collections. We have also added literature on nonstationary estimation and explained the distinct dynamical role of the signed higher-order terms. The section and equation numbers below refer to the revised manuscript. We respectfully request reconsideration on the basis of these clarifications and revisions.

1. State spaces and finite entropy capacity

Concern. The state spaces and entropy capacity are insufficiently specified, and the relation to continuous-state systems and coarse graining is unclear.

Response. We agree that the state-space assumptions and the meaning of capacity should be stated at the outset. The revised Sec. 2.1 defines an N-component process in discrete time, with each component taking values in a fixed finite alphabet $\mathcal{M}_i$. The alphabet entropy capacity is explicitly defined as

$$C_X := \max_{p \text{ on } \mathcal{M}_X} H_p(X) = \ln |\mathcal{M}_X|.$$

This is an alphabet bound, rather than a channel capacity optimized over an input distribution. We also state $H(X'|X) \le H(X') \le C_X$ and distinguish the conditional-entropy margin used in the receiver argument from marginal-entropy saturation.

Finite capacity is part of the model specification. It can describe intrinsically discrete components, as in our logic-gate examples, or labels produced by a fixed finite partition of a continuous state space. In the latter case, the conclusions concern the observed labels and depend on the chosen partition. Coarse graining is therefore one possible source of a finite alphabet, rather than a necessary origin of finite capacity in every model considered here. We explicitly state that Markovianity, determinism, and reversibility must be checked again after coarse graining. The finite-alphabet proofs are not applied directly to unpartitioned continuous variables.

Manuscript changes. Sec. 2.1 now introduces the process and capacity before the information measures; Sec. 5.3 defines the remaining conditional-entropy margin. The alphabet size of receiver P is denoted by q_P, separately from the downstream component Q.

2. Nonstationary ensemble dynamics and estimation from data

Concern. It is unclear whether the time dependence arises from nonstationary distributions or local information values. Reconstructing distributions from time series commonly involves a stationarity assumption.

Response. We acknowledge that stationarity-based analyses have produced important insights into complex systems. For example, Barnett et al. [1] use stationary information measures to distinguish the behavior of global information flow from that of mutual information near an ordering transition. The value of this approach is not in question. Its success, however, does not remove the need to develop theory for nonstationary processes and their transient information dynamics.

The present work uses the reviewer's first interpretation: *ensemble-averaged quantities evaluated with time-dependent joint probability laws.* The initial ensemble and update protocol supply the dynamics. A non-invariant initial law can evolve under a fixed transition rule; switching on the external interaction can also change the subsequent joint laws. Under strict stationarity, the ensemble quantities considered here are indeed constant, while transfer entropy itself can remain nonzero. Local information values along a trajectory provide a different description, which we now identify explicitly with reference to Lizier et al. [2].

We agree that stationarity is often used to justify pooling observations over time in one trajectory. It is not required by the definitions of entropy or transfer entropy. One established alternative reconstructs the relevant distributions from an ensemble of repeated realizations at corresponding times. Gómez-Herrero et al. [3] develop time-resolved estimates on this basis, and Wollstadt et al. [4] implement ensemble transfer-entropy estimation for nonstationary neural data.

Practical analyses also use time windows whose position and duration are selected to resolve changing interactions. Sliding-window effective transfer entropy is an example [5]. Within-window pooling normally entails a local-stationarity approximation. A shorter window provides finer temporal resolution but fewer samples; a longer window can blend different statistical regimes. Analysis-window selection is also important for repeated-trial estimation [4]. Consequently, an adjustable window is an estimation device whose assumptions must be assessed, not a reason to assume that the entire process is stationary.

Our simulations estimate distributions separately at each time across independent realizations. They do not infer a transient distribution by pooling the full trajectory under a global stationarity assumption. We also clarify the scope of the theoretical contribution: the exact identities analyze a specified family of consistent probability laws; they do not claim to solve distribution estimation for arbitrary nonstationary data. We therefore believe that the practical importance of stationary estimation and the development of a dynamical theory for nonstationary ensembles are compatible research directions.

Manuscript changes. The Introduction and Sec. 2.2 now explain this motivation and cite stationary, ensemble, and window-based analyses. Sec. 2.3 distinguishes ensemble transfer entropy from its local values, and Sec. 6 states how the simulations construct time-specific empirical distributions.

3. Starting point, notation, and self-contained information balances

Concern. The manuscript relies on the earlier framework without adequately introducing its notation, quantities, or derivation.

Response. We agree that the identities needed for the present argument should be understandable without first reading the earlier paper [6]. We have therefore reorganized Sec. 2 around the underlying process and the entropy chain rule. Sec. 2.1 specifies the initial ensemble and a transition-kernel description. Sec. 2.3 defines the prime notation and simultaneous time shift, including shifts of conditioning variables. Sec. 2.4 defines ordinary block transfer, the signed multiple-transfer terms, and their correction E.

The derivation in Sec. 2.5 starts with the elementary identity

$$\begin{aligned}\Delta H(X) &= H(X'|X) - H(X|X') \\ &= T_{\mathbf{K}\to X} - T^{\dagger}_{\mathbf{K}\to X} + \alpha_1^{K\to X}, \\ \alpha_1^{K\to X} &:= H(X'|X,K) - H(X|X',K').\end{aligned}$$

Here $\mathbf{K}$ denotes the source block, and the last equality follows by expanding the two conditional mutual informations. Inclusion–exclusion then separates the block contribution into pairwise transfers and E. The manuscript similarly derives the mutual-information balance and obtains both transfer-

entropy balances by taking finite differences of their source-entropy representations. The compact equation system is presented only after these steps.

All residual coefficients are defined in Eq. (21), with an explanation of the conditional uncertainties and finite differences they represent. Although the earlier terminology calls them control variables, their values are determined by the joint laws. The identities do not introduce an autonomous process for information measures with arbitrarily assignable coefficients. The zero-control conditions used later are stated as additional restrictions on a stochastic process.

Manuscript changes. Secs. 2.3–2.5 provide the notation and pairwise derivations. Appendix B retains the higher-order hierarchy and defines its ordered-bipartition index set. The long cycle/source specialization is placed with its derivation in Appendix C, while the local response remains in the main text.

4. Signed multiple transfer and the dynamical role of higher-order terms

Concern. The multiple-transfer quantities are insufficiently motivated, ordinary transfer entropy already applies to groups of variables, and co-information-based quantities can be negative.

Response. We agree with both mathematical points. Ordinary transfer entropy applies to two random-vector blocks and is nonnegative. The multiple-transfer terms in this manuscript are conditional co-informations and can be negative. The revision makes this distinction explicit and does not interpret a negative value as a negative amount of ordinary block transfer.

Their purpose is to retain the exact higher-order corrections required by a componentwise dynamical description. For two sources,

$$T_{(A,B)\to P} = T_{A\to P} + T_{B\to P} - T_{\{A,B\}\to P}.$$

The left side is ordinary block transfer, whereas the final term is a signed correction. In the independent-input XOR example, the two pairwise transfers vanish, the block transfer is $\ln 2$, and the correction is $-\ln 2$. Its negative sign is therefore necessary for the exact identity.

More generally, the entropy representation supplies the recursion

$$I(S \cup \{Z\}|W) = I(S|W) - I(S|W, Z).$$

Taking a simultaneous time difference gives the corresponding recursion for the changes of these quantities. This structure supports inductive arguments for controlling higher-order terms: their time dependence can be tracked, and cancellations can be established under the stated residual conditions. The cycle/source argument uses such a higher-order cancellation to preserve $E = 0$ away from the merging component. We now explain this role and do not claim that recursion alone closes the hierarchy or makes arbitrary higher-order correlations vanish.

PID offers an important interpretive advantage by separating redundant, unique, and synergistic contributions [7]. That distinction is useful when information is interpreted or estimated from data. PID also has applications to information transfer [8]; we do not suggest that it is restricted to estimation or stationary systems. The present choice of quantities serves a different purpose: deriving exact, recursive balance relations for the evolution of information functionals. It does not require selecting a PID redundancy measure, and a signed co-information is not identified with an individual PID contribution.

We therefore respectfully ask that these quantities be reconsidered in their role within a dynamical theory, where their recursive entropy identities are essential, rather than judged solely as candidates for a nonnegative decomposition of informational contributions. Their interpretive limitations and their algebraic usefulness can both be acknowledged.

Manuscript changes. Sec. 2.4 now contrasts block and signed transfer, gives the XOR example and the recursion in Eq. (15), and explains the relationship to PID. Appendix B connects the recursion to the higher-order balance hierarchy; Appendix C states the conditions used for the cancellation through time.

5. Graph vertices, information edges, and causal-parent sets

Concern. The graph and its vertices are ambiguous, and the graph relative to which causal parents are defined is unspecified.

Response. The revision distinguishes three objects. A vertex of the component graph labels a component process $\{\Gamma_i(t)\}_t$. A state value is an element of $\mathcal{M}_i$, whereas a time-unrolled graph would use a separate vertex (i, t) for each component and time. The graph in the manuscript is the component graph.

The structural dependency collections are specified by the update model. Definition 6 calls $K_i \subseteq \{\Gamma_j : j \neq i\}$ sufficient when there are forward and inverse local update relations of the form

$$\Gamma_i' = U_i(\Gamma_i, K_i; t), \qquad \Gamma_i = (U^{-1})_i(\Gamma_i', K_i'; t+1).$$

The term causal-parent set is used in this explicit structural sense. Such a collection can include variables required for the inverse update, need not be minimal, and is not a parent set inferred from pairwise transfer entropy. The full collection of other components is sufficient under deterministic reversible dynamics; a smaller sufficient collection may be selected and held fixed.

For labels $\mathcal{P}_i := \{j : \Gamma_j \in K_i\}$, the chosen structural graph records the relation $j \in \mathcal{P}_i$. The information-flow graph is then defined separately by

$$G_T(t) = (V, E_T(t)), \qquad E_T(t) = \{(j, i) : j \in \mathcal{P}_i,\ T_{j\to i}(t) > 0\}.$$

An update dependency need not produce positive transfer for every ensemble. Conversely, pairwise predictive dependence alone is not treated as an intervention-based causal statement. Fixed support means that the positive-edge set is unchanged over the stated time interval, although its weights may vary. The proof also states the sufficient-neighborhood convention used to compare the forward and reversed-transfer graphs, and imposes its non-synergy condition on those chosen collections.

Manuscript changes. Sec. 2.7 gives the sufficient-dependency definition and the update relations. Sec. 3 identifies the vertices and defines the structural and information-flow graphs, fixed support, and information circulation.

References

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